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Optical design and analysis of a high-speed triple galvanometer laser 3D scanning system

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ABSTRACT

In this paper, a triple galvanometer laser three-dimensions (3D) scanning system (TGLSS) was proposed. The system consisted of three galvanometers and two parabolic mirrors, with rapid movement of the laser focus in 3D space by rotations of the three galvanometers. The optical principle of the TGLSS was analyzed, and the method of designing TGLSS by geometrical optics and Abbe's sine condition was given in detail. Scanning range and focusing quality were analyzed by optical simulations. An experiment was carried out to verify the feasibility of TGLSS. The implications of TGLSS for reflective dynamic focusing device (RDFD)-based laser system and their impact on technologically relevant applications were discussed. The great potential for enhancing the speed of advanced laser processing in 3D space was offered by TGLSS.

1. Introduction

Laser scanning has a wide range of applications, such as manufacturing [1–4], texture processing [5–7], cutting [8–10], cleaning [11–14], and imaging [15,16]. The speed and precision of laser scanning affect the efficiency and quality of the laser processes [17–19], which are the most critical concerns during the processing. For curved parts, line width and scanning continuity could affect the dimensional accuracy or functional characteristics of the finished product [20,21]. Due to its simple structure, high speed, and high precision, a two-axis galvanometer is commonly used for two-dimensional (2D) laser scanning [22,23]. The scanning speed on the plane perpendicular to the optical axis (XY plane) can reach 10 m/s [24]. With the complexity of scanned workpieces and requirements, the need for laser scanning has been put forward in three dimensions (3D). Achieving a rapid movement of the focus in the axial (Z-) direction is the main challenge in laser scanning from 2D to 3D. Conventional Z-focusing control methods are mainly based on the mechanical movement of optical components [25–27]. The Z-focusing speed is significantly limited by the moving speed of the optical components and their attachments. Consequently, the scanning speed of the laser focus in the Z-direction is three orders of magnitude slower than in the X- and Y-directions [28] and the 3D scanning speed of laser systems is limited.

Recently, advanced Z-focusing optical elements have emerged to meet the needs of industry and science. Methods based on electrowetting [29,30], shape-variable lenses [31–33] or Alvarez lenses [34–36] allow for rapid movement of the Z-focus. However, the curvatures of the lenses are difficult to control precisely, and the 3D scanning speed of the laser systems based on the above varifocal optics is also limited to the order of 1 m/s due to the slow response rate [27]. Reflective dynamic focusing devices (RDFDs) use two highly dynamic properties galvanometers to convert angular motion into axial motion, increasing the laser scanning speed in the Z-direction [37–39]. However, the optical principle of RDFDs is not well discussed, the relationship between structure and aberration has not been analyzed, and no systematic design method has been proposed. In addition, an RDFD-based laser 3D scanning system requires a total of four galvanometers, resulting in high costs, large volume and requiring four control channels.

In this paper, the optical principle for RDFDs was analyzed and the design method was proposed. A triple galvanometer laser 3D scanning system (TGLSS) was proposed based on the method, with 3D scanning speed up to 10 m/s. Specifically, as shown in Fig. 1, two parabolic

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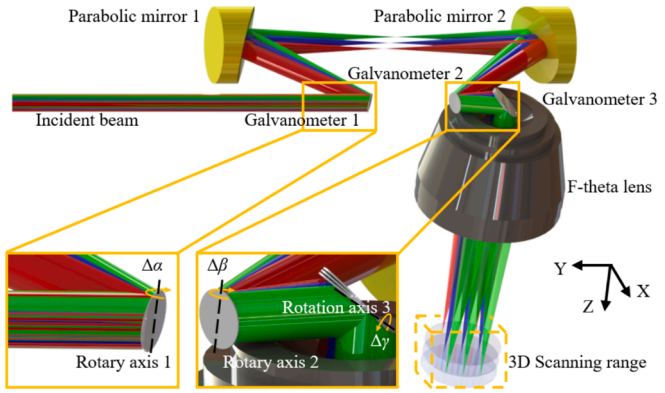


Fig. 1. A conceptual design of a TGLSS.

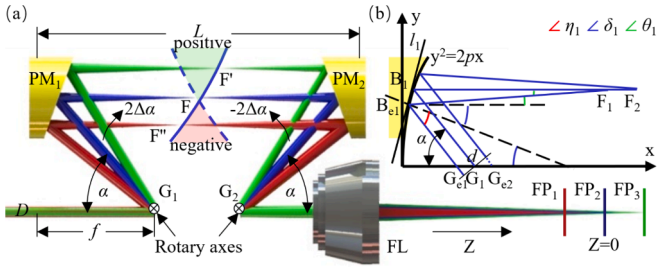


Fig. 2. Z-focus module of the TGLSS (a) Schematic diagram; (b) part of the optical path.

mirrors (PM_1, PM_2), three galvanometers (G_1, G_2 and G_3) and an F-theta lens (FL) were utilized. The rotation angle $\Delta\alpha$ of the G_1 and the rotation angle $\Delta\beta$ of the G_2 together determined the Z-coordinate of the focus, in addition, the rotation angle $\Delta\beta$ of the G_2 also determined the X-coordinate of the focus. The rotation angle $\Delta\gamma$ of the G_3 determined the Y-coordinate of the focus. Different 3D scanning ranges could be realized by different combinations of the rotation angles of the three galvanometers. Since three identical galvanometers had the same dynamic properties, it was possible to manipulate the laser focus with high speed, high precision, and synchronization. In addition, the TGLSS was less expensive, smaller, and had fewer control channels compared to the conventional RDFD-based laser 3D scanning system.

2. Optical design

2.1. Design and analysis of Z-focus module

The design scheme diagram of the Z-focus module of a TGLSS and part of the geometrical optical path model were shown in Fig. 2. The D was the laser beam diameter. The PM_1 and PM_2 were two parabolic mirrors with the same focal length f and off-axis angle α. G_1 and G_2 were the galvanometers placed at the focal points of PM_1 and PM_2, whose rotary axes passed through the focal points of PMs. The rotation angles of G_1 and G_2 were equivalent, and the rotation directions were opposite, denoted by $\Delta\alpha$ and $-\Delta\alpha$ respectively. So that the deflection angles of the optical paths were $2\Delta\alpha$ and $-2\Delta\alpha$ respectively. Since the edge rays from G_1 were not pass through the focal point of PM_1, the edge rays exiting PM_1 were not parallel but converge between PM_1 and PM_2 to form the primary focus F.

When the galvanometers were rotated, the path of F formed a tilted image plane between PM_1 and PM_2, as shown by the blue solid line in Fig. 2(a). When F located in the middle position of PM_1 and PM_2, the Z-focus module was a symmetric structure, which satisfied the Abbe sine condition [40], in which case the focus had no coma and spherical aberrations. For the α corresponding to this F, the distance L between PM_1 and PM_2 in this case was L_{Best}. When G_1 and G_2 were rotated around α corresponding to L_{Best}, the object plane of PM_2 and the image plane of PM_1 do not coincide, as shown by the blue dashed line in Fig. 2(a). When the primary focus F' was in the positive focusing region, the focus was shifted in the + Z direction, and when the primary focus F'' was in the negative focusing region, the focus was tuned in the −Z direction and focused by the FL, as shown in the focal planes (FPs) FP_1, FP_2 and FP_3 in Fig. 2(a).

Fig. 2(b) represented the optical path of the laser from G_1 to the primary focus F. For the rays emanating from the center G_1 and edge G_{e1}, could be calculated using the following equations:

$$\begin{cases} \tan\alpha = y_{B_1}/(f - x_{B_1}) \\ y_{B_1}^2 = 2px_{B_1} = 4fx_{B_1} \\ y_{B_1} = (-2f + \sqrt{4f^2 + 4f^2\tan^2\alpha})/\tan\alpha \end{cases} \tag{1}$$

$$\begin{cases} \tan\alpha = y_{B_{e1}}/(f - d/2 - x_{B_{e1}}) \\ y_{B_{e1}}^2 = 2px_{B_{e1}} = 4fx_{B_{e1}} \\ y_{B_{e1}} = (-2f + \sqrt{4f^2 - 2df\tan^2\alpha + 4f^2\tan^2\alpha})/\tan\alpha \end{cases} \tag{2}$$

where B_1 and B_{e1} were the intersection of the chief ray with PM_1 and the intersection of the edge ray and PM_1, respectively. The focal parameter of the parabola was p, and the laser diameter D was projected on the x-axis at a distance d. The angular relationship between the chief ray and the edge ray satisfies:

$$\begin{aligned} \alpha &= \eta_1 + \delta_1 \\ \theta_1 &= \eta_1 - \delta_1 \end{aligned} \tag{3}$$

$$\begin{cases} k_{l_1} = \tan(\pi/2 - \delta_1) \\ k_{l_1} = 2f/y_{B_{e1}} \end{cases} \tag{4}$$

$$\theta_1 = \alpha - \pi + 2\arctan(k_{l_1}) \tag{5}$$

where η_1 was the angle of incidence of the edge ray, θ_1 was the angle between the chief ray and the edge ray, and δ_1 was the angle between the normal at point B_{e1} and the x-axis. k_{l1} was the slope of the tangent line at B_{e1}. The intersection F_1 of the edge ray from B_{e1} with the chief ray could be calculated:

$$\begin{aligned} x_{F_1} &= x_{B_{e1}} + (y_{B_1} - y_{B_{e1}})/\tan\theta_1 \\ &= f - d/2 + (2f - m_1)/\tan^2\alpha + \frac{\sqrt{4f^2 + 4f^2\tan^2\alpha} - m_1}{\tan\alpha\tan(\alpha + 2\arctan(2f\tan\alpha/(m_1 - 2f)))} \end{aligned} \tag{6}$$

$$m_1 = \sqrt{4f^2 - 2df\tan^2\alpha + 4f^2\tan^2\alpha}$$

Similarly, the intersection F_2 of the edge ray from B_{e2} with the chief ray could be calculated:

$$\begin{aligned} x_{F_2} &= f + d/2 + (2f \\ &\quad - m_2)/\tan^2\alpha + \frac{\sqrt{4f^2 + 4f^2\tan^2\alpha} - m_2}{\tan\alpha\tan(\alpha + 2\arctan(2f\tan\alpha/(m_2 - 2f)))} \end{aligned} \tag{7}$$

$$m_2 = \sqrt{4f^2 + 2df\tan^2\alpha + 4f^2\tan^2\alpha}$$

When the distance L was equal to the sum of x_{F1} and x_{F2}, the PM_1 and PM_2 were symmetric with respect to point F. According to Abbe sine condition, the distance L in this case was L_{Best} at no aberration could be obtained to satisfy:

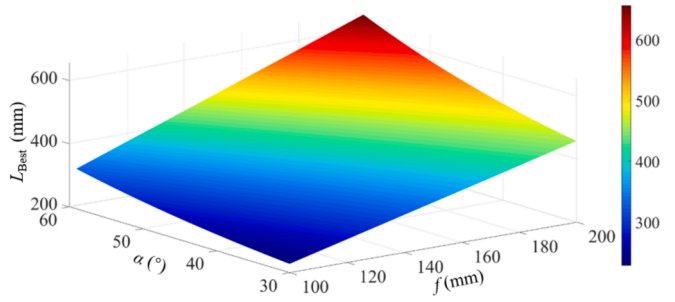


Fig. 3. Results of the theoretical analysis: variations of L_{Best} with α and f (D=8mm).

$$L_{Best} = 2f + 4f/\tan^2\alpha + \sum_{i=1}^{2}\left(\frac{\sqrt{4f^2 + 4f^2\tan^2\alpha} - m_i}{\tan\alpha\tan(\alpha + 2\arctan(2\alpha/(m_i - 2f)))} - m_i/\tan^2\alpha\right) \quad (8)$$

The relationship between the L_{Best} and the parameters α and f calculated by Eq. (8) was shown in Fig. 3. It could be seen that the L_{Best} increased with α or f increased. This was due to that the location of the more distant primary focus F determined by a larger α or f.

When the diffraction of the ray was considered, the Strehl ratio (STRH) reflected the effect of the laser light on the focusing effect in terms of energy [41]. Therefore, the STRH was used to evaluate the focusing effect within the scanning range, taking the value of 0 to 1, with the closer to 1 indicating better focusing. A STRH $\geq$ 0.8 was generally considered to correspond to diffraction-limited performance and effective processing. For STRH$\leq$0.3, the behavior of the STRH became uncorrelated to the processing quality due to large aberrations [41]. The STRH at the focus of the ZEMAX simulation was shown in Fig. 4 when f and L were varied, and Fig. 4(a)~(c) were for α = 30°, 45°, and 60°, respectively. The simulation results showed that for a given value of α, there was a region of best focusing effect (the yellow part of the figure), and this region varies with f and L.

The red curves in Fig. 4(a)~(c) showed the L_{Best} calculated by Eq. (8). With $\alpha = 30°$, the L_{Best} increased from 228.708 mm to 457.432 mm when f increased from 100 mm to 200 mm. With α = 45°, the L_{Best} increased from 268.606 mm to 537.247 mm when f increased from 100 mm to 200 mm. With $\alpha = 60°$, the L_{Best} increased from 333.293 mm to 666.647 mm when f increased from 100 mm to 200 mm. The L_{Best} results calculated by Eq. (8) were all within the optimal STRH region for the ZEMAX simulation, which proved the accuracy of the equation. The detailed data were shown in Table 1.

After determining the L_{Best} based on the data in Table 1, the Z-focus module of the TGLSS could be determined. With FL (focal length f_{FL} = 100 mm) focusing, the variation of the Z-coordinate of the focal point (Z) with $\Delta\alpha$ of G_1 ($-\Delta\alpha$ of G_2) and f of PMs (α = 30°, 45°, 60°) could be calculated. The specific curves were shown in Fig. 5. When the $\Delta\alpha$ of G_1 varied from −2° to 2° ($-\Delta\alpha$ of G_2 varied from 2° to −2°), Z increased

Table 1
Theoretically calculated values of L_{Best} (mm).

	f = 100 mm	f = 125 mm	f = 150 mm	f = 175 mm	f = 200 mm
α = 30°	228.708	285.890	343.071	400.252	457.432
α = 45°	268.606	335.768	402.928	470.088	537.247
α = 60°	333.293	416.635	499.973	583.316	666.647

with increasing $\Delta\alpha$. The magnitude of this change decreased with increasing f and increased with increasing α.

The ΔZ was used to represent the Z-focus range when $\Delta\alpha$ varied between −2° and 2°, as shown in Table 2.

2.2. Design and analysis of X- and Y- focus model

In the proposed Z-focus module of the TGLSS, by rotating G_1 and G_2 located at the focal points of PM_1 and PM_2, the movement of the focus along the Z-direction could be realized to determine the focal plane of the TGLSS. By setting a G_3 at a distance l after the G_2 with a rotational axis orthogonal to the G_2 rotational axis, the G_2 and G_3 could form the X- and Y- focus module of the TGLSS. The detailed process was shown in Fig. 6.

The relationship between the position of the focus in the determined Z-plane and the rotation angles was shown in Fig. 6. It could be seen that the focus position in the X-direction could be changed by further rotating the rotation axis R_2 of G_2, and the focus position in the Y-direction could be changed by rotating the rotation axis R_3 of G_3, and the rotation angles θ_x of G_2 and θ_y of G_3 could be derived as

$$\theta_x = \frac{1}{2}\arctan(x/(l + \sqrt{h^2 + y^2})) \quad (9)$$

$$\theta_y = \frac{1}{2}\arctan(y/h) \quad (10)$$

where x and y were the coordinates of laser focus P position, l was the distance between the center of G_2 and G_3, and h was the distance between the center of G_3 and the laser focus P along the Z-direction.

2.3. Parameter analysis and scanning simulation

Combined with 2.1 and 2.2, the rotation $\Delta\alpha$ of G_1 was determined by the Z-coordinate of the focus P, the rotation $\Delta\beta$ of G_2 was determined by the Z- and X-coordinates of the focus P, and the rotation $\Delta\gamma$ of G_3 was determined by the Y-coordinate of the focus P. The rotation of G_2 satisfied the following equation:

$$\begin{cases} \Delta\alpha = \Delta\alpha(Z) \\ \Delta\beta = -\Delta\alpha + \theta_x \\ \Delta\gamma = \theta_y \end{cases} \quad (11)$$

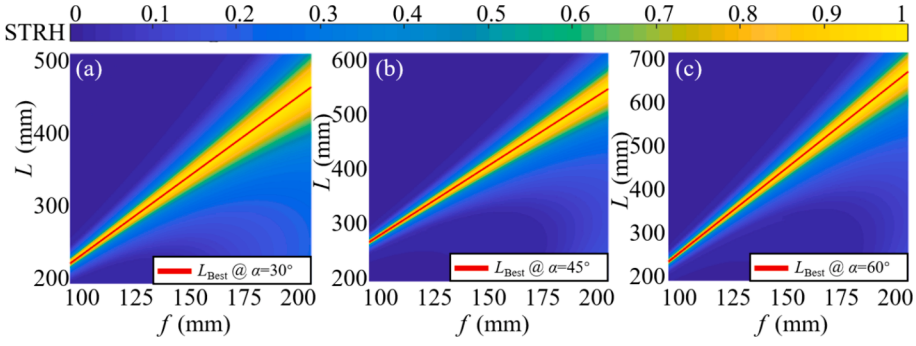


Fig. 4. Theoretical and ZEMAX simulation analysis results: variation of L_{Best} with α and f (D=8 mm).

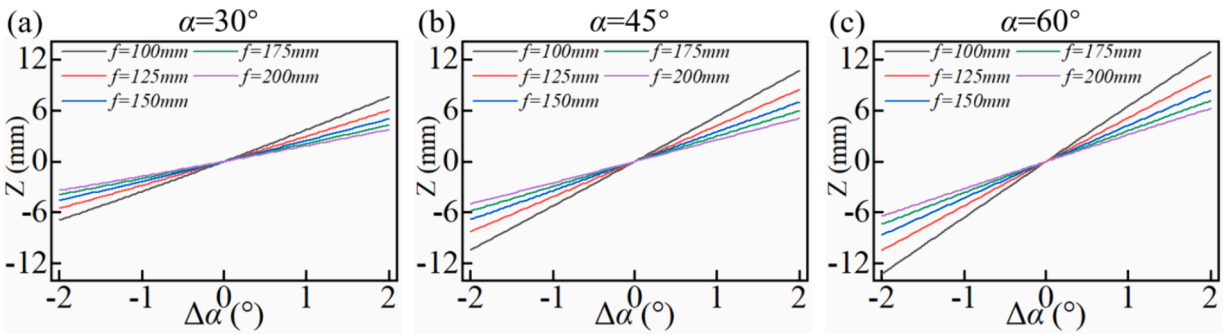


Fig. 5. Variations of the Z-coordinate of the focal point with $\Delta\alpha$ of G_1 ($-\Delta\alpha$ of G_2) and f of PMs ($\alpha = 30°$, $45°$, $60°$), and the focal length f_{FL} of FL=100 mm.

Table 2
Simulation values of ΔZ (mm).

	$f = 100$ mm	$f = 125$ mm	$f = 150$ mm	$f = 175$ mm	$f = 200$ mm
$\alpha = 30°$	14.499	11.550	9.598	8.210	7.173
$\alpha = 45°$	21.105	16.716	13.837	11.803	10.093
$\alpha = 60°$	26.134	20.658	17.077	14.553	12.678

Fig. 7 showed the simulation of the 3D scanning range and focusing effect of the TGLSS. The volume of the cube represented the 3D scanning range, and the color bar represented the STRH value of the focusing point. $\Delta\alpha$ ranged from $-2°$ to $+2°$, θ_x from $-5°$ to $+5°$, and θ_y from $-5°$ to $+5°$. At off-axis angle $\alpha = 30°$, $45°$, and $60°$, the 3D scanning range decreased with increasing parabolic focal length f, and when $f = 100$ mm, 125 mm, 150 mm, 175 mm, 200 mm, the 3D scanning range increased with the increasing α.

The variations of focus uniformity with respect to α and f were shown in Fig. 7. At $\alpha = 30°$, $45°$, and $60°$, the focus uniformity improved in the scanning range with increasing f. When $f = 100$ mm, 125 mm, 150 mm, 175 mm, 200 mm, the focus uniformity decreased in the scanning range with the increase of α. The variation of the effective processing (STRH≥0.8) points as a percentage of the 3D scanning range was showed in Table 3.

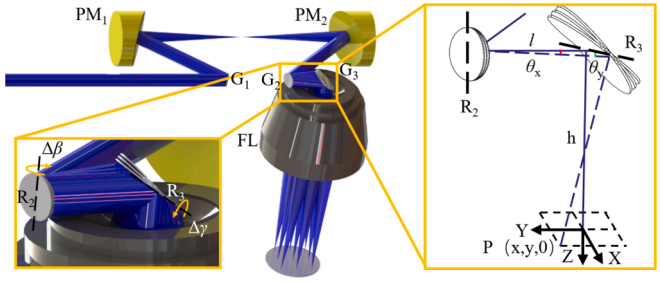


Fig. 6. X- and Y- focus module of the TGLSS (a) Schematic diagram; (b) part of the optical path.

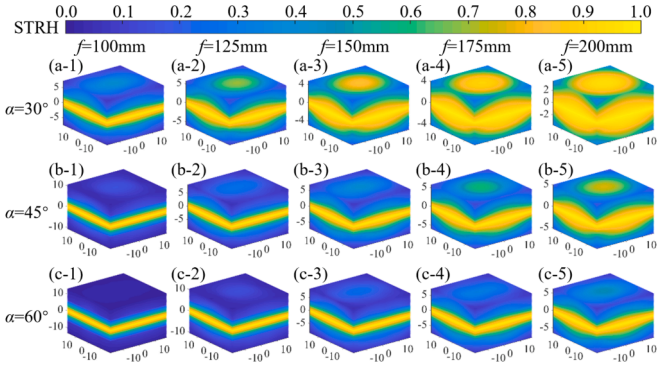


Fig. 7. Simulation of TGLSS 3D scanning range and effect ($\Delta\alpha$ from $-2°$ to $+2°$, θx from $-5°$ to $+5°$, θy from $-5°$ to $+5°$).

3. Verification of rapid 3D scanning

3.1. Experiment setup

An experiment was carried out to verify the 3D scanning capability of the TGLSS, and the experimental setup is shown in Fig. 8. A nanosecond fiber laser (YDFLP-100-M8-M−R, JPT) with a 3.0 mm (expanded to 8 mm, for observing the experimental results more dramatically) beam diameter was adopted for the experiment. The wavelength was 1064 nm, the beam quality factor M^2 was 1.4, and the maximum average power was 100 W. The focal length f of the off-axis parabolic mirrors (35588, Edmund) was 152.4 mm, and the off-axis angle α was 45°, corresponding to the initial state of G_1 and G_2. The G_1, G_2 and G_3 were galvanometers (ExtraScan II 14, Hans), which could be rotated by ± 12.5°. An F-theta lens with a focal length f_{FL} of 100 mm was used to focus the laser. The distance of PM_1 and PM_2 was 410.729 mm, calculated by Eq. (8). An XYZ stage was placed to move the samples. A 45° block was placed on the XYZ stage for tilting an aluminum plate 1060 with a size of 50 × 50 × 1 mm^3, used for the experiment. Three different 3D scanning modes were selected to verify the 3D scanning capability of the TGLSS: horizontal, tilted, and vertical. The laser scanning path was a 20 × 20 mm^2 rectangle, and each scan was repeated 10 times using the above three scanning modes, all with lines spacing of 250 μ m. The laser processing effect was used as the criterion for verifying the 3D scanning capability. The experimental laser energy was 100 μJ, the laser frequency was 100 kHz, and the 3D scanning speed was 10 m/s. A laser scanning confocal microscope (VK-X1100, Keyence) was used to observe and compare the scanning effect.

3.2. Results and discussion

The microscopic observations of 3D processing on a 45° inclined

Table 3
Effective processing (STRH≥0.8) points as a percentage of the 3D scanning range.

	$f = 100$ mm	$f = 125$ mm	$f = 150$ mm	$f = 175$ mm	$f = 200$ mm
$\alpha = 30°$	22.4 %	39.3 %	56.8 %	71.2 %	81.0 %
$\alpha = 45°$	12.4 %	16.8 %	26.2 %	38.3 %	50.6 %
$\alpha = 60°$	10.9 %	12.6 %	16.25 %	23.1 %	32.8 %

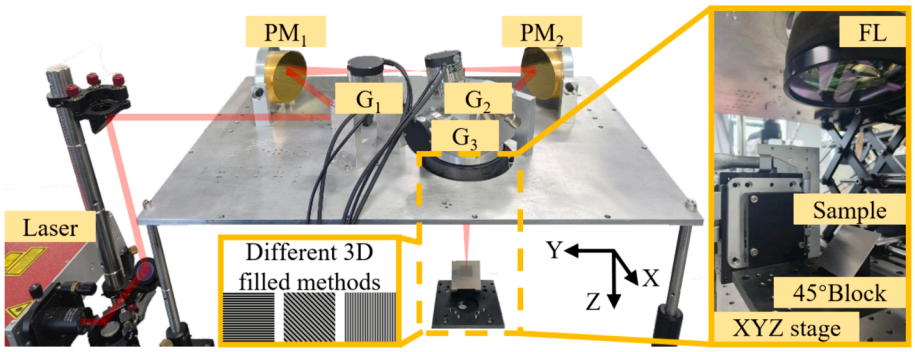


Fig. 8. The experiment setup of the TGLSS.

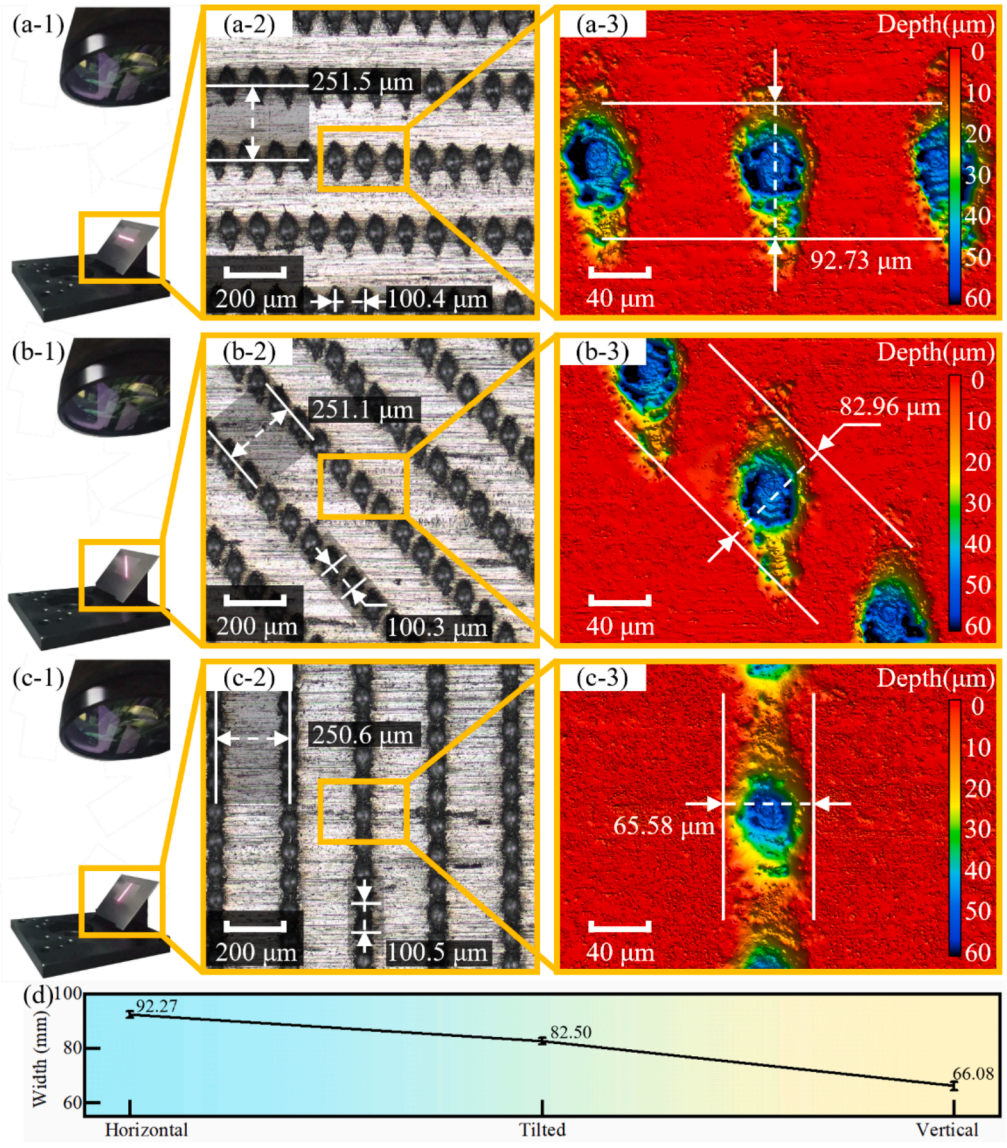


Fig. 9. The scanning microscope observations of the system in a 45° inclined plane. (a) horizontal scanning; (b) tilted scanning; (c) vertical scanning; (d) variation of line width with scanning mode.

plane were shown in Fig. 9. The horizontal scanning mode on a 45° inclined plane was shown in Fig. 9(a-1). The observation of this mode at 10 × magnification was showed in Fig. 9(a-2), and the line spacing was 251.5 μm and the focus spacing was 100.4 μm. In the horizontal scanning mode, there was a completely unfilled area between consecutive scanned points. The observation of this mode at 50 × magnification was showed in Fig. 9(a-3), since the laser focused on the 45° inclined plane, the spots were no longer circular, but the projections (ellipses) of circles on the inclined plane. Consequently, the line width of the scan was 92.73 μm as shown in Fig. 9(a-3).

The tilted scanning mode on a 45° inclined plane was shown in Fig. 9 (b-1). The observation of this mode at 10 × magnification was showed in Fig. 9(b-2), and the line spacing was 251.1 μm and the focus spacing was 100.3 μm. It was showed that the connections between successive scanned points were tighter in the tilted scanning mode than in the horizontal scanning mode. The observation of this mode at 50 × magnification was showed in Fig. 9(b-3), and the line width of the scan was 82.96 μm as shown in Fig. 9(b-3), which was narrower than that of the horizontal mode.

The vertical scanning mode on a 45° inclined plane was shown in Fig. 9(c-1). The observation of this mode at 10 × magnification was showed in Fig. 9(c-2), and the line spacing was 250.6 μm and the focus spacing was 100.5 μm. In the vertical scanning mode, the spots were connected to each other. As shown in Fig. 9 (c-3), the line width of the vertical mode was 65.58 μm at 50 × magnification, which was narrower than that of the tilted mode.

The lines spacing shown in Fig. 9(a-2), (b-2), and (c-2) were about 250 μm, corresponding to the preset line spacing, and the focus spacing were about 100 μm, corresponding to the preset 100 kHz laser frequency and 10 m/s 3D scanning speed. The depth of the scanning spot shown in Fig. 9(a-3), (b-3), and (c-3) did not change under the three different 3D scanning modes, because the spot overlaps were not high and the depth depended only on the single pulse laser ablation. In addition, with the scanning direction changed from the short axis to the long axis of the ellipse projected by the focus, the line width became narrower and the connection became more successive, as shown in Fig. 9(a-3), (b-3), (c-3), and (d). It showed the capability of TGLSS to adjust line width and scanning continuity in 3D space, which could help achieve better efficiency and quality of processing [42–48].

4. Conclusion

In conclusion, the optical principle and design method of TGLSS were proposed and discussed, and the optical structure of TGLSS with no aberration was given by geometrical optics calculation and Abbe's sine condition. The design method could be applied to other similar reflective dynamic focusing devices (RDFDs). The simulations were performed to analyze the 3D scanning range and effect of TGLSS under different parameters, where the axial focusing range ΔZ was varied between 7.173 mm and 26.134 mm. In this experimentally verification, the improvement in axial scanning speed (10 m/s) enabled different laser 3D scanning modes with adjustable line width and scanning continuity. Other laser processing applications such as manufacturing, texture processing, cutting and cleaning will also benefit from this development.

CRediT authorship contribution statement

Yi Liu: Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tianting Chen:** Writing – review & editing. **Yaqing Qiao:** Writing – review & editing, Visualization, Conceptualization. **Yuhang Ding:** Software. **Ziqiang Dan:** Data curation. **Wangrong Lu:** Software. **Haoran Ma:** Investigation. **Guiyang Yang:** Validation. **Chenfei Zhang:** Resources. **Wei Xiong:** Writing – review & editing, Supervision. **Leimin Deng:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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